

CUSTOMER	: GOODSCIENCE SDN. BHD.
DESCRIPTION	: MECOBAL 500MCG TABLET
MEASUREMENT	: 160MM (W) X 240MM (H)
MATERIAL	: PDSM 050

SHEET SIZE	: NIL	CLOUR :
PROCESS	: -	BLACK
FINISHING	: NIL	
BOARD OUTING	: NIL	

MATERIAL CODE:	Attn : JANNAH
PM-00156/REV04	From : MALA
	I hereby verify & confirm that this draft is the final copy. Any subsequent changes of the design and content of the end product will not be borne by THUNDER PRINT SDN. BHD.
	Customer Signature & Date

16.7.2025

MECOBAL 500MCG TABLET

CONTENT

Each Tablet contains mecobalamin 500mcg

Excipients: Lactose, Cornstarch, Povidone K90, Polysorbate 80, Microcrystalline cellulose, Talc, Magnesium stearate, Hypromellose E5, Hypromellose E15, Titanium dioxide & PEG 6000.

DESCRIPTION

A round biconvex tablet coated with white film.

PHARMACODYNAMICS

ATC Code: B03BA05

Pharmacotherapeutic group: Vitamin B12 (cyanocobalamin and analogues)

Mecobalamin accelerates the metabolic pathways of nuclei acids, proteins, and lipids by acting as a coenzyme in methionine synthesis. In particular, it was found to be involved in the synthesis of thymidine from deoxyuridine and to accelerate the synthesis of DNA and RNA. Mecobalamin was also found to accelerate the synthesis of lecithin, a major component of the myelin sheath.

Mecobalamin is efficiently transferred to nervous tissues and improves metabolic disorders. Mecobalamin has also been demonstrated to help maintain axonal function by promoting the synthesis of structural proteins and normalizing the transport velocity of these proteins.

In animal studies, mecobalamin was neuropathologically and electrophysiologically shown to inhibit the onset of nerve degeneration due to nerve disturbances caused by adriamycin and vincristine and experimental diabetes caused by streptozotocin. Mecobalamin was also found to be effective as steroids for the recovery from the paralysis when evaluated based on the winking reflex, induced electromyography and histological evaluation.

Mecobalamin accelerates the synthesis of nucleic acids in bone marrow, as well as the maturation and division of the erythroblast, resulting in an increase in production of erythrocytes.

PHARMACOKINETICS

It was shown in healthy adult subjects that dose-dependent peak plasma concentrations were achieved after 3 hours following oral administration of single dose of 120 and 1500mcg of mecobalamin. From the cumulative amount of total B12 recovered in the urine 24 hours after oral administration, about 40-80% were excreted within the first 8 hours.

For repeated oral dose administration (1500mcg), the plasma concentration was increased for the first 4 weeks after administration, reaching a value twice as high as the initial concentration. Then, it gradually increased up to about 280% of the initial value at 12th week of dosing. Four weeks after discontinuation of last dosing, the plasma concentration declined to approximately 180% of the initial plasma level.

Vitamin B12 is extensively bound to specific plasma proteins called transcobalamins; transcobalamin II appears to be involved in the rapid transport of the cobalamins to tissues. Vitamin B12 is stored in the liver. Vitamin B12 diffuses across the placenta and also appears in breast milk.

Elimination Half-life:

12.5 hrs (single-dose oral administration; calculated from the average of 24-48 hour values)

INDICATION

Peripheral neuropathies. Megaloblastic anemia due to vitamin B12 deficiency.

RECOMMENDED DOSAGE

For oral administration

Adult: 1 tablet 3 times per day (1500 mcg of mecobalamin daily) or as directed by physician.

CONTRAINDICATIONS

Mecobal 500mcg Tablet is contraindicated in patients with hypersensitivity to mecobalamin or any of its excipients.

WARNING / PRECAUTIONS

This product should not be used aimlessly for more than one month unless it is effective.

Vitamin B12 should, if possible, not be given to patients with suspected vitamin B12 deficiency without first confirming the diagnosis. Where it is desirable to start therapy immediately, combined treatment for both deficiencies may be started once suitable samples have been taken to permit diagnosis of the deficiency, and the patient converted to the appropriate treatment once the cause of the anaemia is known. Regular monitoring of the blood is advisable.

Although the haematological symptoms of B12 deficiency and folate deficiency are similar, it is important to distinguish between them since the use of folate alone in B12-deficient megaloblastic anaemia can improve haematological symptoms without preventing aggravation of accompanying neurological symptoms, and may lead to severe nervous system sequelae such as subacute combined degeneration of the spinal cord. Use of doses

greater than 10 micrograms daily may produce a haematological response in patients with folate deficiency and indiscriminate use may mask the precise diagnosis. Conversely, folate may mask vitamin B12 deficiency.

Precautions concerning use

Administration: Mecobalamin is susceptible to photolysis. Light decreases the content of mecobalamin and tablets may change colour (e.g., turn reddish) with exposure to moisture. Therefore, this product should be used promptly after the package is opened, and caution should be taken so as not to expose the tablets/capsules to light/moisture.

Other Precautions

The prolonged use of larger doses of mecobalamin is not recommended for patients whose occupation requires the handling of mercury or mercury compounds.

DRUG INTERACTION

Absorption of vitamin B12 from the gastrointestinal tract may be reduced by neomycin, aminosalicylic acid, histamine H2-antagonists, omeprazole, and colchicine.

Serum concentrations may be decreased by use of oral contraceptives.

Many of these interactions are unlikely to be of clinical significance but should be taken into account when performing assays for blood concentrations.

PREGNANCY AND LACTATION

Pregnancy

There are no data available for mecobalamin to be used in pregnant women.

Lactation

There are no data available for mecobalamin to be used in lactating women. However, since vitamin B12 is distributed into breast milk, The American Academy of Pediatrics considers its use to be usually compatible with breast feeding.

SIDE EFFECTS

Gastrointestinal Disorders: Nausea/vomiting, diarrhea

General Disorders and Administration Site Conditions: Chills, fever, malaise

Metabolism and Nutrition Disorders: Anorexia

Nervous System Disorders: Dizziness, tremor, headache

Renal and Urinary Disorders: Chromaturia

Skin and Subcutaneous Tissue Disorder: Itching, acneform, bullous eruptions, rash; In the event of such symptoms, treatment should be discontinued.

Others: Paraesthesia, hot sensation

SYMPTOM & TREATMENT OF OVERDOSE

Mecobalamin is well tolerable and no known toxicity. Treatment is unlikely to be needed in cases of overdose.

EFFECTS ON ABILITY FOR DRIVE AND USE MACHINE

Mecobalamin has no known effect on the ability to drive or operate machine.

PACKING/ PACK SIZES

Available in Alu-Alu blister pack of 10's in packaging of 100's and 500's per box.

STORAGE CONDITION

Do not store above 30°C.

Protect from moisture and light.

Keep out of reach of children.

Jauhi daripada kanak-kanak

Shelf Life: Please refer to the outer box label.

NAME AND ADDRESS OF MANUFACTURERS

Packed and released by:

GoodScience Sdn Bhd

No. 7, Jalan KPK 4/3,

Kawasan Perindustrian Kundang, Kundang Jaya,

48020 Rawang, Selangor, Malaysia

Manufactured by:

T.O Chemicals (1979) Ltd.

Bangkok 10310, Thailand.

PRODUCT REGISTRATION HOLDER

GoodScience Sdn Bhd

No. 7, Jalan KPK 4/3,

Kawasan Perindustrian Kundang, Kundang Jaya,

48020 Rawang, Selangor, Malaysia

Revised Date 16-July-2025

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160mm